AI-Enabled Developer

Module D & E – Building and Retrieval concepts

Level 1: Foundations · Topics B20–B21

Two short bridge modules, shown as worked examples: a real tool-calling transcript with the security boundary marked, and the same question answered ungrounded, grounded, and refused.

2	45	2	10
topics	minutes	concept diagrams	role profiles

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LEVEL 1: FOUNDATIONS

Module D & E at a glance

Two single-topic modules, 45 minutes, both prerequisites for Security. Neither should be cut.

B20	**What tool use actually means** Worked example · full width	25 min
B21	**Grounding and citations** Worked example · full width	20 min

Role chips on every topic slide: FS Full-stack · BE Backend · FE Frontend · QA · DO DevOps/SRE · DE Data Eng · DS Data Sci/ML · PB Product/BA · SEC Security · EM Tech Lead

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What tool use actually means

OUTCOME Explain that the model emits a request and your code executes it — and point at where the security boundary sits.

- You declare the tools. The model **requests** a call. **Your code decides whether to honour it.**
- In the transcript, the model asked to refund £500. Your code checked auth, validated, and gated it to human approval.
- **The model never touched the database** — it emitted a request and read back only what you returned.
- Every tool argument is untrusted input. The model chose it — sometimes an attacker chose it **through** the model.

A REAL TOOL CALL — WHO EXECUTES EACH LINE

YOUR CODE	`declares get_order(id), refund(id, amount)`
MODEL	`requests: refund(id="A-8821", amount=500)`
YOUR CODE	`auth ok? this user may refund? amount ≤ order?`
YOUR CODE	`gate: refund > £100 needs human confirm → HOLD`
YOUR CODE	`returns: {"status":"pending_approval"}`
MODEL	`"I've queued a £5 refund for approval."`

THE TWO ORANGE LINES

are the only place a decision is made. Your auth, your validation, your gate.

The model asked for £500. Your code decided it needed approval.

The model never touched a database — it emitted a request and read a result.

EXERCISE Trace a tool-calling transcript and mark exactly where the security boundary sits.

ROLES: **FS** **BE** FE QA **DO** DE DS PB **SEC** EM

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Grounding and citations

OUTCOME Explain why grounding beats instructing the model not to hallucinate.

- Three real answers to real questions: one invented, one grounded and cited, one correctly refused.
- The ungrounded answer sounds identical in confidence to the grounded one — that is exactly the danger.
- **No citation, no claim.** The grounded answer points at `returns_policy.md`; you can check it, and so can the user.
- **It must be able to refuse.** The out-of-corpus question gets "I don't have that" — a designed output, not a default.

THE SAME QUESTION — UNGROUNDED, GROUNDED, AND REFUSED

UNGROUNDED

```
Q: refund window for
   damaged goods?
A: Typically 30 days,
   though it may vary.

(invented — no source)
```

Confident. Unverifiable.

GROUNDED + CITED

```
Q: refund window for
   damaged goods?
A: 30 days from
   delivery [1].

[1] returns_policy.md
```

Every claim points at a source.

OUT OF CORPUS

```
Q: what's your CEO's
   home address?
A: I don't have that
   in the provided
   documents.
(refusal, by design)
```

Refusal is a built feature.

EXERCISE Ask the same question with and without the source document in context; compare and check the citations.

ROLES: FS | BE | FE | QA | DO | DE | DS | PB | SEC | EM

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PLANNING REFERENCE

Module D & E exercises

Exercises live here rather than on each topic slide. Each is repeated in the speaker notes of its topic.

ID	Topic	Exercise	Min
B20	**What tool use actually means**	Trace a tool-calling transcript and mark exactly where the security boundary sits.	25
B21	**Grounding and citations**	Ask the same question with and without the source document in context; compare and check the citations.	20

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END OF MODULE D & E

What they can do now — and what comes next

AFTER MODULE D & E

- Explain that the model requests a call and their own code executes it.
- Point at where the security boundary sits in a tool-calling flow.
- Treat every tool argument, and every model output, as untrusted input.
- Explain why grounding beats instructing a model not to hallucinate.
- Say why a system that cannot refuse is a liability.

NEXT: MODULE F — SECURITY

Indirect injection, least privilege and credentials, and data handling.

Everything in Module F assumes step three of the flow you just saw. It opens with a live demo of an agent exfiltrating a credential because a document told it to — and it only lands because the room now knows where the action actually happened.

The model is not an actor in your system. It is a very persuasive suggestion engine.

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